

# Digital Electronics Question and Answers

## Topic 1

### Number system & Codes :

Number System – 1

Number System – 2

Arithmetic Operation

1's, 2's, 9's & 10's Complements – 1

Binary Coded Decimal(BCD)

## Digital Electronics Question & Answers

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### Number System & Codes

#### Number System – 1

1. Any signed negative binary number is recognised by its \_\_\_\_\_

- a) MSB
- b) LSB
- c) Byte
- d) Nibble

**Answer: a**

**Explanation:** Any negative number is recognized by its MSB (Most Significant Bit). If it's 1, then it's negative, else if it's 0, then positive.

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2. The parameter through which 16 distinct values can be represented is known as \_\_\_\_\_

- a) Bit
- b) Byte
- c) Word
- d) Nibble

**Answer: c**

**Explanation:** It can be represented up to 16 different values with the help of a Word. Nibble is a combination of four bits and Byte is a combination of 8 bits. It is "word" that is said to be a collection of 16-bits on most of the systems.

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3. If the decimal number is a fraction then its binary equivalent is obtained by \_\_\_\_\_ the number continuously by 2.

- a) Dividing
- b) Multiplying
- c) Adding
- d) Subtracting

**Answer: b**

**Explanation:** On multiplying the decimal number continuously by 2, the binary equivalent is obtained by the collection of the integer part. However, if it's an integer, then it's binary equivalent is determined by dividing the number by 2 and collecting the remainders.

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4. The representation of octal number (532.2)<sub>8</sub> in decimal is \_\_\_\_\_

- a) (346.25)<sub>10</sub>

- b)  $(532.864)_{10}$
- c)  $(340.67)_{10}$
- d)  $(531.668)_{10}$

**Answer: a**

**Explanation:** Octal to Decimal conversion is obtained by multiplying 8 to the power of base index along with the value at that index position.

$$(532.2)_8 = 5 * 8^2 + 3 * 8^1 + 2 * 8^0 + 2 * 8^{-1} = (346.25)_{10}$$


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5. The decimal equivalent of the binary number  $(1011.011)_2$  is \_\_\_\_\_

- a)  $(11.375)_{10}$
- b)  $(10.123)_{10}$
- c)  $(11.175)_{10}$
- d)  $(9.23)_{10}$

**Answer: a**

**Explanation:** Binary to Decimal conversion is obtained by multiplying 2 to the power of base index along with the value at that index position.

$$1 * 2^3 + 0 * 2^2 + 1 * 2^1 + 1 * 2^0 + 0 * 2^{-1} + 1 * 2^{-2} + 1 * 2^{-3} = (11.375)_{10}$$

Hence,  $(1011.011)_2 = (11.375)_{10}$

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6. An important drawback of binary system is \_\_\_\_\_

- a) It requires very large string of 1's and 0's to represent a decimal number
- b) It requires sparingly small string of 1's and 0's to represent a decimal number
- c) It requires large string of 1's and small string of 0's to represent a decimal number
- d) It requires small string of 1's and large string of 0's to represent a decimal number

**Answer: a**

**Explanation:** The most vital drawback of binary system is that it requires very large string of 1's and 0's to represent a decimal number. Hence, Hexadecimal systems are used by processors for calculation purposes as it compresses the long binary strings into small parts.

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7. The decimal equivalent of the octal number  $(645)_8$  is \_\_\_\_\_

- a)  $(450)_{10}$
- b)  $(451)_{10}$
- c)  $(421)_{10}$
- d)  $(501)_{10}$

**Answer: c**

**Explanation:** Octal to Decimal conversion is obtained by multiplying 8 to the power of base index along with the value at that index position.

The decimal equivalent of the octal number  $(645)_8$  is  $6 * 8^2 + 4 * 8^1 + 5 * 8^0 = 6 * 64 + 4 * 8 + 5 = 384 + 32 + 5 = (421)_{10}$ .

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8. The largest two digit hexadecimal number is \_\_\_\_\_

- a) (FE)<sub>16</sub>
- b) (FD)<sub>16</sub>
- c) (FF)<sub>16</sub>
- d) (EF)<sub>16</sub>

**Answer: c**

**Explanation:** (FE)<sub>16</sub> is 254 in decimal system, while (FD)<sub>16</sub> is 253. (EF)<sub>16</sub> is 239 in decimal system. And, (FF)<sub>16</sub> is 255. Thus, The largest two-digit hexadecimal number is (FF)<sub>16</sub>.

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9. Representation of hexadecimal number (6DE)<sub>16</sub> in decimal:

- a)  $6 * 16^2 + 13 * 16^1 + 14 * 16^0$
- b)  $6 * 16^2 + 12 * 16^1 + 13 * 16^0$
- c)  $6 * 16^2 + 11 * 16^1 + 14 * 16^0$
- d)  $6 * 16^2 + 14 * 16^1 + 15 * 16^0$

**Answer: a**

**Explanation:** Hexadecimal to Decimal conversion is obtained by multiplying 16 to the power of base index along with the value at that index position.

In hexadecimal number D & E represents 13 & 14 respectively.

So,  $6DE = 6 * 16^2 + 13 * 16^1 + 14 * 16^0$ .

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10. The quantity of double word is \_\_\_\_\_

- a) 16 bits
- b) 32 bits
- c) 4 bits
- d) 8 bits

**Answer: b**

**Explanation:** One word means 16 bits, Thus, the quantity of double word is 32 bits.

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## Number System – 2

1. The given hexadecimal number (1E.53)<sub>16</sub> is equivalent to \_\_\_\_\_

- a) (35.684)<sub>8</sub>
- b) (36.246)<sub>8</sub>
- c) (34.340)<sub>8</sub>
- d) (35.599)<sub>8</sub>

**Answer: b**

**Explanation:** First, the hexadecimal number is converted to its equivalent binary form, by writing the binary equivalent of each digit in form of 4 bits. Then, the binary equivalent bits are grouped in terms of 3 bits and then for each of the 3-bits, the respective digit is written. Thus, the octal equivalent is obtained.

$$\begin{aligned}
 (1E.53)_{16} &= (0001\ 1110.0101\ 0011)_2 \\
 &= (00011110.01010011)_2 \\
 &= (011110.010100110)_2 \\
 &= (011\ 110.010\ 100\ 110)_2 \\
 &= (36.246)_8.
 \end{aligned}$$


---

2. The octal number  $(651.124)_8$  is equivalent to \_\_\_\_\_

- a)  $(1A9.2A)_{16}$
- b)  $(1B0.10)_{16}$
- c)  $(1A8.A3)_{16}$
- d)  $(1B0.B0)_{16}$

**Answer: a**

**Explanation:** First, the octal number is converted to its equivalent binary form, by writing the binary equivalent of each digit in form of 3 bits. Then, the binary equivalent bits are grouped in terms of 4 bits and then for each of the 4-bits, the respective digit is written. Thus, the hexadecimal equivalent is obtained.

$$\begin{aligned}
 (651.124)_8 &= (110\ 101\ 001.001\ 010\ 100)_2 \\
 &= (110101001.001010100)_2 \\
 &= (0001\ 1010\ 1001.0010\ 1010)_2 \\
 &= (1A9.2A)_{16}.
 \end{aligned}$$


---

3. The octal equivalent of the decimal number  $(417)_{10}$  is \_\_\_\_\_

- a)  $(641)_8$
- b)  $(619)_8$
- c)  $(640)_8$
- d)  $(598)_8$

**Answer: a**

**Explanation:** Octal equivalent of decimal number is obtained by dividing the number by 8 and collecting the remainders in reverse order.

$$8 \mid 417$$

$$8 \mid 52 - 1$$

$$8 \mid 6 - 4$$

$$\text{So, } (417)_{10} = (641)_8.$$


---

4. Convert the hexadecimal number  $(1E2)_{16}$  to decimal.

- a) 480
- b) 483
- c) 482
- d) 484

**Answer: c**

**Explanation:** Hexadecimal to Decimal conversion is obtained by multiplying 16 to the power of base index along with the value at that index position.

$$(1E2)_{16} = 1 * 16^2 + 14 * 16^1 + 2 * 16^0 \text{ (Since, E = 14)}$$

$$= 256 + 224 + 2 = (482)_{10}.$$


---

5.  $(170)_{10}$  is equivalent to \_\_\_\_\_

- a)  $(FD)_{16}$
- b)  $(DF)_{16}$
- c)  $(AA)_{16}$
- d)  $(AF)_{16}$

**Answer: c**

**Explanation:** Hexadecimal equivalent of decimal number is obtained by dividing the number by 16 and collecting the remainders in reverse order.

$$16 \mid 170$$

$$16 \mid 10 - 10$$

$$\text{Hence, } (170)_{10} = (AA)_{16}.$$


---

6. Convert  $(214)_8$  into decimal.

- a)  $(140)_{10}$
- b)  $(141)_{10}$
- c)  $(142)_{10}$
- d)  $(130)_{10}$

**Answer: a**

**Explanation:** Octal to Decimal conversion is obtained by multiplying 8 to the power of base index along with the value at that index position.

$$(214)_8 = 2 * 8^2 + 1 * 8^1 + 4 * 8^0$$

$$= 128 + 8 + 4 = (140)_{10}.$$


---

7. Convert  $(0.345)_{10}$  into an octal number.

- a)  $(0.16050)_8$
- b)  $(0.26050)_8$
- c)  $(0.19450)_8$
- d)  $(0.24040)_8$

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**Answer: b**

**Explanation:** Converting decimal fraction into octal number is achieved by multiplying the fraction part by 8 everytime and collecting the integer part of the result, unless the result is 1.

$$0.345 \times 8 = 2.76 \text{ 2}$$

$$0.760 \times 8 = 6.08 \text{ 6}$$

$$00.08 \times 8 = 0.64 \text{ 0}$$

$$0.640 \times 8 = 5.12 \text{ 5}$$

$$0.120 \times 8 = 0.96 \text{ 0}$$

$$\text{So, } (0.345)_{10} = (0.26050)_8.$$


---

8. Convert the binary number  $(01011.1011)_2$  into decimal.

a)  $(11.6875)_{10}$

b)  $(11.5874)_{10}$

c)  $(10.9876)_{10}$

d)  $(10.7893)_{10}$

**Answer: a**

**Explanation:** Binary to Decimal conversion is obtained by multiplying 2 to the power of base index along with the value at that index position.

$$(01011)_2 = 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 11$$

$$(1011)_2 = 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4} = 0.6875$$

$$\text{So, } (01011.1011)_2 = (11.6875)_{10}.$$


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9. Octal to binary conversion:  $(24)_8 = ?$

a)  $(111101)_2$

b)  $(010100)_2$

c)  $(111100)_2$

d)  $(101010)_2$

**Answer: b**

**Explanation:** Each digit of the octal number is expressed in terms of group of 3 bits. Thus, the binary equivalent of the octal number is obtained.

$$(24)_8 = (010100)_2.$$


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10. Convert binary to octal:  $(110110001010)_2 = ?$

a)  $(5512)_8$

b)  $(6612)_8$

c)  $(4532)_8$

d)  $(6745)_8$

**Answer: b**

**Explanation:** The binary equivalent is segregated into groups of 3 bits, starting from left. And then for each group, the respective digit is written. Thus, the octal equivalent is obtained.

$$(110110001010)_2 = (6612)_8.$$


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## Arithmetic Operation

1. What is the addition of the binary numbers 11011011010 and 010100101?

- a) 0111001000
- b) 1100110110
- c) 11101111111
- d) 10011010011

**Answer: c**

**Explanation:** The rules for Binary Addition are :

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 0 \text{ ( Carry 1)}$$

1

1 1 0 1 1 0 1 1 0 1 0

+ 0 0 0 1 0 1 0 0 1 0 1

---

1 1 1 0 1 1 1 1 1 1 1

---

2. Perform binary addition: 101101 + 011011 = ?

- a) 011010
- b) 1010100
- c) 101110
- d) 1001000

**Answer: d**

**Explanation:** The rules for Binary Addition are :

$$0 + 0 = 0$$

$$0 + 1 = 1$$



$$1 + 0 = 1$$

$$1 + 1 = 0 \text{ ( Carry 1)}$$

$$1\ 1\ 1\ 1\ 1\ 1$$

$$1\ 0\ 1\ 1\ 0\ 1$$

$$+ 0\ 1\ 1\ 0\ 1\ 1$$

---


$$1\ 0\ 0\ 1\ 0\ 0\ 0$$


---

**Therefore, the addition of 101101 + 011011 = 1001000.**

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3. Perform binary subtraction:  $101111 - 010101 = ?$

- a) 100100
- b) 010101
- c) 011010
- d) 011001

**Answer: c**

**Explanation: The rules for Binary Subtraction are :**

$$0 - 0 = 0$$

$$0 - 1 = 1 \text{ ( Borrow 1)}$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

$$1\ 0\ 1\ 1\ 1\ 1$$

$$- 0\ 1\ 0\ 1\ 0\ 1$$

---


$$0\ 1\ 1\ 0\ 1\ 0$$


---

**Therefore, The subtraction of  $101111 - 010101 = 011010$ .**

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4. Binary subtraction of  $100101 - 011110$  is?

- a) 000111
- b) 111000
- c) 010101
- d) 101010

**Answer: a**

**Explanation: The rules for Binary Subtraction are :**

$$0 - 0 = 0$$

$$0 - 1 = 1 \text{ (Borrow 1)}$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

$$100101$$

$$-011110$$

---

$$000111$$

---

**Therefore, The subtraction of  $100101 - 011110 = 000111$ .**

5. Perform multiplication of the binary numbers:  $01001 \times 01011 = ?$

a) 001100011

b) 110011100

c) 010100110

d) 101010111

**Answer: a**

**Explanation: The rules for binary multiplication are:**

$$0 * 0 = 0$$

$$0 * 1 = 0$$

$$1 * 0 = 0$$

$$1 * 1 = 1$$

$$01001$$

$$\times 01011$$

---

$$01001$$

$$010010$$

$$0000000$$

$$01001000$$

$$000000000$$

---

**0 0 1 1 0 0 0 1 1**

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**Therefore,  $01001 \times 01011 = 001100011$ .**

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6.  $100101 \times 0110 = ?$

- a) 1011001111
- b) 0100110011
- c) 1011111110
- d) 011011110

**Answer: d**

**Explanation: The rules for binary multiplication are:**

**$0 * 0 = 0$**

**$0 * 1 = 0$**

**$1 * 0 = 0$**

**$1 * 1 = 1$**

**1 0 0 1 0 1**

**x    0 1 1 0**

---

**0 0 0 0 0 0**

**1 0 0 1 0 1 0**

**1 0 0 1 0 1 0 0**

**0 0 0 0 0 0 0 0 0**

---

**0 1 1 0 1 1 1 1 0**

---

**Therefore,  $100101 \times 0110 = 011011110$ .**

---

7. On multiplication of (10.10) and (01.01), we get \_\_\_\_\_

- a) 101.0010
- b) 0010.101
- c) 011.0010
- d) 110.0011

**Answer: c**

**Explanation: The rules for binary multiplication are:**

$$0 * 0 = 0$$

$$0 * 1 = 0$$

$$1 * 0 = 0$$

$$1 * 1 = 1$$

$$\begin{array}{r}
 10.10 \\
 \times 01.01 \\
 \hline
 1010 \\
 00000 \\
 101000 \\
 0000000 \\
 \hline
 011.0010
 \end{array}$$

**Therefore,  $10.10 \times 01.01 = 011.0010$ .**

8. Divide the binary numbers:  $111101 \div 1001$  and find the remainder.

a) 0010

b) 1010

c) 1100

d) 0111

**Answer: d**

**Explanation: Binary Division is accomplished using long division method.**

$$\begin{array}{r}
 1001 \overline{) 111101} \text{ (11} \\
 \underline{1001} \phantom{00} \\
 01100 \\
 \underline{1001} \phantom{00} \\
 \phantom{0}100
 \end{array}$$

**0 1 1 1****Therefore, the remainder of  $111101 \div 1001 = 0111$ .**

---

9. Divide the binary number (011010000) by (0101) and find the quotient.

a) 100011

b) 101001

c) 110010

d) 010001

**Answer: b****Explanation:****0 1 0 1 ) 0 1 1 0 1 0 0 0 0 ( 0 1 0 1 1 1****0 0 0 0**

---

**0 1 1 0 1****0 0 1 0 1**

---

**0 1 0 0 0 0****0 0 0 0 0 0**

---

**1 0 0 0 0****0 0 1 0 1**

---

**0 1 0 1 1 0****0 0 0 1 0 1**

---

**1 0 0 0 1 0****0 0 0 1 0 1**

---

**1 1 1 0 1 0**

$$\begin{array}{r}
 000101 \\
 \hline
 10101 \\
 00101 \\
 \hline
 10000
 \end{array}$$

Therefore, the quotient of  $011010000 \div 1001 = 101001$ .

---

10. Binary subtraction of  $101101 - 001011 = ?$

- a) 100010
- b) 010110
- c) 110101
- d) 101100

**Answer: a**

**Explanation:** The rules for binary subtraction are:

$$0 - 0 = 0$$

$$0 - 1 = 1 \text{ (Borrow 1)}$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

$$101101$$

$$-001011$$

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$$100010$$

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Therefore, the subtraction of  $101101 - 001011 = 100010$ .

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## 1's, 2's, 9's & 10's Complements – 1

1. 1's complement of 1011101 is \_\_\_\_\_

- a) 0101110
- b) 1001101
- c) 0100010
- d) 1100101

**Answer: c**

**Explanation:** 1's complement of a binary number is obtained by reversing the binary bits. All the 1's to 0's and 0's to 1's.

**Thus, 1's complement of 1011101 = 0100010.**

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2. 2's complement of 11001011 is \_\_\_\_\_

- a) 01010111
- b) 11010100
- c) 00110101
- d) 11100010

View Answer

**Answer: c**

**Explanation:** 2's complement of a binary number is obtained by finding the 1's complement of the number and then adding 1 to it.

**2's complement of 11001011 = 00110100 + 1 = 00110101.**

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3. On subtracting  $(01010)_2$  from  $(11110)_2$  using 1's complement, we get \_\_\_\_\_

- a) 01001
- b) 11010
- c) 10101
- d) 10100

**Answer: d**

**Explanation:** Steps For Subtraction using 1's complement are:

-> 1's complement of the subtrahend is determined and added to the minuend.

-> If the result has a carry, then it is dropped and 1 is added to the last bit of the result.

-> Else, if there is no carry, then 1's complement of the result is found out and a '-' sign precedes the result.

|                                |           |
|--------------------------------|-----------|
|                                | 1 1 1     |
| Minuend -                      | 1 1 1 1 0 |
| 1's complement of subtrahend - | 1 0 1 0 1 |

|              |   |           |
|--------------|---|-----------|
| Carry over - | 1 | 1 0 0 1 1 |
|              |   | 1         |
|              |   | 1 0 1 0 0 |

4. On subtracting  $(010110)_2$  from  $(1011001)_2$  using 2's complement, we get \_\_\_\_\_

- a) 0111001
- b) 1100101
- c) 0110110
- d) 1000011

**Answer: d**

**Explanation: Steps For Subtraction using 2's complement are:**

-> 2's complement of the subtrahend is determined and added to the minuend.

-> If the result has a carry, then it is dropped and the result is positive.

-> Else, if there is no carry, then 2's complement of the result is found out and a '-' sign precedes the result.

|                                |               |       |
|--------------------------------|---------------|-------|
| 1's complement of subtrahend - | 1 1 0 1 0 0 1 |       |
|                                |               | _____ |
|                                |               | 1 1 1 |
| Minuend -                      | 1 0 1 1 0 0 1 |       |
| 2's complement of subtrahend - | 1 1 0 1 0 1 0 |       |
|                                |               | _____ |
| Carry over - 1                 | 1 0 0 0 0 1 1 |       |

**Answer: 1000011**

5. On subtracting  $(001100)_2$  from  $(101001)_2$  using 2's complement, we get \_\_\_\_\_

- a) 1101100
- b) 011101
- c) 11010101
- d) 11010111

**Answer: b**

**Explanation: Steps For Subtraction using 2's complement are:**

-> 2's complement of the subtrahend is determined and added to the minuend.

-> If the result has a carry, then it is dropped and the result is positive.

-> Else, if there is no carry, then 2's complement of the result is found out and a '-' sign precedes the result.

|                                |             |       |
|--------------------------------|-------------|-------|
| 1's complement of subtrahend - | 1 1 0 0 1 1 |       |
|                                |             | _____ |



|                                |                    |
|--------------------------------|--------------------|
| Minuend -                      | <u>1 0 1 0 0 1</u> |
| 2's complement of subtrahend - | 1 1 0 1 0 0        |
| Carry over - 1                 | <u>0 1 1 1 0 1</u> |

**Answer: 011101**

6. On addition of 28 and 18 using 2's complement, we get \_\_\_\_\_

- a) 00101110
- b) 0101110
- c) 00101111
- d) 1001111

**Answer: b**

**Explanation: Steps for Binary Addition Using 2's complement:**

-> The binary equivalent of the two numbers are obtained and added using the rules of binary addition.

|          |               |
|----------|---------------|
| Augend - | 0 0 1 1 1 0 0 |
| Addend - | 0 0 1 0 0 1 0 |

0 1 0 1 1 1 0

**Answer: 0 1 0 1 1 1 0**

7. On addition of +38 and -20 using 2's complement, we get \_\_\_\_\_

- a) 11110001
- b) 100001110
- c) 010010
- d) 110101011

**Answer: c**

**Explanation: Steps for Binary Addition Using 2's complement:**

-> The 2's complement of the addend is found out and added to the first number.

-> The result is the 2's complement of the sum obtained.

|                               |               |
|-------------------------------|---------------|
| Augend -                      | 0 1 0 0 1 1 0 |
| 2's Complement of Subtrahend: | 1 1 0 1 1 0 0 |

---

1    0 0 1 0 0 1 0

**Answer: 0 1 0 0 1 0**

---

8. On addition of -46 and +28 using 2's complement, we get \_\_\_\_\_

- a) -10010
- b) -00101
- c) 01011
- d) 0100101

**Answer: a**

**Explanation:** The BCD form is written of the two given numbers, in their signed form. After which, normal binary addition is performed.

**Augend is 28 and Subtrahend is -46.**

|                                      |                                 |
|--------------------------------------|---------------------------------|
| <b>Augend -</b>                      | <b>0 0 1 1 1 0 0    ....(a)</b> |
| <b>2's Complement of Subtrahend:</b> | <b>1 0 1 0 0 1 0    ....(b)</b> |

---

|                             |                      |
|-----------------------------|----------------------|
| <b>Addiing (a) and (b):</b> | <b>1 1 0 1 1 1 0</b> |
|-----------------------------|----------------------|

Since, there is no carry, so answer will be negative  
and 2's complement of the above result is determined.

|          |                      |
|----------|----------------------|
|          | <b>0 0 1 0 0 0 1</b> |
| <b>+</b> | <b>1</b>             |

---

**0 0 1 0 0 1 0**

**Answer: - 1 0 0 1 0**

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9. On addition of -33 and -40 using 2's complement, we get \_\_\_\_\_

- a) 1001110
- b) -110101
- c) 0110001
- d) -1001001

**Answer: d**

**Explanation:** The BCD form is written of the two given numbers, in their signed form. After

which, normal binary addition is performed.

Augend is -40 and Subtrahend is -33.

|                               |                          |
|-------------------------------|--------------------------|
| Augend -                      | 1 0 1 0 0 0 0 1 .....(a) |
| 2's Complement of Subtrahend: | 1 1 0 1 1 0 0 1 .....(b) |

|                     |                   |
|---------------------|-------------------|
| Adding (a) and (b): | 1 0 1 0 0 1 0 0 0 |
|---------------------|-------------------|

Since, there is no carry, so answer will be negative

and 2's complement of the above result is determined.

|   |               |
|---|---------------|
|   | 1 0 0 1 0 0 0 |
| + | 1             |
|   | 1 0 0 1 0 0 1 |

**Answer: -1001001**

10. On subtracting +28 from +29 using 2's complement, we get \_\_\_\_\_

- a) 11111010
- b) 111111001
- c) 100001
- d) 1

**Answer: d**

**Explanation: Steps For Subtraction using 2's complement are:**

-> 2's complement of the subtrahend is determined and added to the minuend.

-> If the result has a carry, then it is dropped and the result is positive.

-> Else, if there is no carry, then 2's complement of the result is found out and a '-' sign precedes the result.

|                                |             |
|--------------------------------|-------------|
| 1's complement of subtrahend - | 1 0 0 0 1 1 |
| Minuend -                      | 0 1 1 1 0 1 |
| 2's complement of subtrahend - | 1 0 0 1 0 0 |

|              |             |
|--------------|-------------|
| Carry over - | 1 0 0 0 0 1 |
|--------------|-------------|

**Answer: 000001 = 1**

11. On subtracting +28 from +29 using 2's complement, we get \_\_\_\_\_

- a) 11111010
- b) 111111001
- c) 100001
- d) 1

**Answer: d**

**Explanation: Steps For Subtraction using 2's complement are:**

-> 2's complement of the subtrahend is determined and added to the minuend.

-> If the result has a carry, then it is dropped and the result is positive.

-> Else, if there is no carry, then 2's complement of the result is found out and a '-' sign precedes the result.

1's complement of subtrahend -    1 0 0 0 1 1

Minuend -                    0 1 1 1 0 1

2's complement of subtrahend -    1 0 0 1 0 0

---

Carry over -    1   0 0 0 0 1

**Answer: 000001 = 1**

12. If the number of bits in the sum exceeds the number of bits in each added numbers, it results in \_\_\_\_\_

- a) Successor
- b) Overflow
- c) Underflow
- d) Predecessor

**Answer: c**

**Explanation: If the number of bits in the sum exceeds the number of bits in each added numbers, it results in overflow and is also known as excess-one. In case of any arithmetic operation, if the result has less number of bits than the operands, then it is known as underflow condition.**

13. An overflow is a \_\_\_\_\_

- a) Hardware problem
- b) Software problem
- c) User input problem
- d) Input Output Error

**Answer: b**

**Explanation: An overflow is a software problem which occurs when the processor cannot handle the result properly when it produces an out of the range output.**

---

14. An overflow occurs in \_\_\_\_\_

- a) MSD position
- b) LSD position
- c) Middle position
- d) Signed Bit

**Answer: a**

**Explanation: An overflow occurs at the Most Significant Digit position. It occurs when the processor cannot handle the result properly when it produces an out of the range output.**

---

15. Logic circuitry is used to detect \_\_\_\_\_

- a) Underflow
- b) MSD
- c) Overflow
- d) LSD

**Answer: c**

**Explanation: To check the overflow logic circuitry is used in each case. Overflow occurs when the processor cannot handle the result properly when it produces an out of the range output.**

---

15. 1's complement can be easily obtained by using \_\_\_\_\_

- a) Comparator
- b) Inverter
- c) Adder
- d) Subtractor

**Answer: b**

**Explanation: With the help of inverter the 1's complement is easily obtained. Since, during the operation of 1's complement 1 is converted into 0 and vice-versa and this is well suited for the inverter.**

---

16. The advantage of 2's complement system is that \_\_\_\_\_

- a) Only one arithmetic operation is required
- b) Two arithmetic operations are required
- c) No arithmetic operations are required
- d) Different Arithmetic operations are required

**Answer: a**

**Explanation:** The advantage of 2's complement is that only one arithmetic operation is required for 2's complement's operation and that is only addition. Just by adding a 1 bit to 1's complement, we get 2's complement.

---

17. The 1's complements requires \_\_\_\_\_

- a) One operation
- b) Two operations
- c) Three operations
- d) Combined Operations

**Answer: a**

**Explanation:** Only one operation is required for 1's complement operation. This includes only inversion of 1's to 0's and 0's to 1's.

---

18. Which one is used for logical manipulations?

- a) 2's complement
- b) 9's complement
- c) 1's complement
- d) 10's complement

**Answer: c**

**Explanation:** For logical manipulations, 1's complement is used, as all logical operations take place with binary numbers.

---

19. For arithmetic operations only \_\_\_\_\_

- a) 1's complement is used
- b) 2's complement
- c) 10's complement
- d) 9's complement

**Answer: b**

**Explanation:** Only 2's complement is used for arithmetic operations, as it is more fast.

---

20. The addition of +19 and +43 results as \_\_\_\_\_ in 2's complement system.

- a) 11001010
- b) 101011010
- c) 00101010
- d) 111110

**Answer: d**

**Explanation:** The decimal numbers are converted to their respective binary equivalent and then the binary addition rules are applied.

---

## Binary Coded Decimal(BCD)

1. Binary coded decimal is a combination of \_\_\_\_\_

- a) Two binary digits
- b) Three binary digits
- c) Four binary digits
- d) Five binary digits

View Answer

**Answer: c**

**Explanation: Binary coded decimal is a combination of 4 binary digits. For example-8421.**

---

2. The decimal number 10 is represented in its BCD form as \_\_\_\_\_

- a) 10100000
- b) 01010111
- c) 00010000
- d) 00101011

**Answer: c**

**Explanation: The decimal number 10 is represented in its BCD form as 0001 0000, in accordance to 8421 for each of the two digits.**

---

3. Add the two BCD numbers:  $1001 + 0100 = ?$

- a) 10101111
- b) 01010000
- c) 00010011
- d) 00101011

**Answer: c**

**Explanation: Firstly, Add the 1001 and 0100. We get 1101 as output but it's not in BCD form. So, we add 0110 (i.e. 6) with 1101. As a result we get 10011 and it's BCD form is 0001 0011.**

---

4. Carry out BCD subtraction for  $(68) - (61)$  using 10's complement method.

- a) 00000111
- b) 01110000
- c) 100000111
- d) 011111000

**Answer: a**

**Explanation: First the two numbers are converted into their respective BCD form using 8421 sequence. Then binary subtraction is carried out.**

---

5. Code is a symbolic representation of \_\_\_\_\_ information.

- a) Continuous
- b) Discrete
- c) Analog
- d) Both continuous and discrete

**Answer: b**

**Explanation:** Code is a symbolic representation of discrete information, which may be present in the form of numbers, letters or physical quantities. Mostly, it is represented using a particular number system like decimal or binary and such like.

---

6. When numbers, letters or words are represented by a special group of symbols, this process is called \_\_\_\_\_

- a) Decoding
- b) Encoding
- c) Digitizing
- d) Inverting

**Answer: b**

**Explanation:** When numbers, letters or words are represented by a special group of symbols, this process is called encoding. Encoding in the sense of fetching the codes or words in a computer. It is done to secure the transmission of information.

---

7. A three digit decimal number requires \_\_\_\_\_ for representation in the conventional BCD format.

- a) 3 bits
- b) 6 bits
- c) 12 bits
- d) 24 bits

**Answer: c**

**Explanation:** The number of bits needed to represent a given decimal number is always greater than the number of bits required for a straight binary encoding of the same. Hence, a three digit decimal number requires 12 bits for representation in BCD format.

---

8. How many bits would be required to encode decimal numbers 0 to 9999 in straight binary codes?

- a) 12
- b) 14
- c) 16
- d) 18



**Answer: b**

**Explanation:** Total number of decimals to be represented = 10000 =  $10^4 = 2^n$  (where n is the number of bits required) =  $2^{13.29}$ . Therefore, the number of bits required for straight binary encoding = 14.

---

9. The excess-3 code for 597 is given by \_\_\_\_\_

- a) 100011001010
- b) 100010100111
- c) 010110010111
- d) 010110101101

**Answer: a**

**Explanation:** The addition of '3' to each digit yields the three new digits '8', '12' and '10'. Hence, the corresponding four-bit binary equivalents are 100011001010, in accordance to 8421 format.

---

10. The decimal equivalent of the excess-3 number 110010100011.01110101 is \_\_\_\_\_

- a) 970.42
- b) 1253.75
- c) 861.75
- d) 1132.87

**Answer: a**

**Explanation:** The conversion of binary numbers into digits '1100', '1010', '0011', '0111' and '0101' gives '12', '5', '3', '7' and '5' respectively. Hence, the decimal number is 970.42.

---

1. Give an example for Non-weighted code?

**Answer:**

Unlike weighted code, non-weighted codes will not any weights. For example, Excess-3 code and Gray code. So the numbers that are represented using non-weighted code can not be directly converted to decimal equivalents.

---

2. What is weighted code? Give example.

**Answer:**

The weighted code will have a fixed weight for each position. For example, in normal binary system, the decimal equivalent can be obtained by multiplying the position value with position weight and adding them together.

---

3. Define: (a) bit (b) nibble (c) byte (d) word

**Answer:**

**a. Bit: Binary digit (Either logic-1 or logic-0) b. Nibble: 4-bits together is called a nibble c. Byte: 8 bits or 2 nibbles d. Word: 16 bits or 2 bytes.**

---

4. What is the key feature of Excess-3 code?

**Answer:**

**Self-complementing: The 9's complement of an excess 3 number can be obtained simply by replacing its 1's with 0's and 0's with 1's.**

---

5. A number N has 'n' digits in a r-radix number system. What is its (r-1)'s complement and r's complement?

**Answer:**

**$(r^n - 1) - N$  ,  $(r^n - N)$**

---

6. Represent 45 in the number systems (a) binary (b) BCD (c) Excess-3

**Answer:**

**(a) 101101 (b) To get BCD: Represent each digit separately in binary 0100 0101 (c) Excess-3: Add 3 to each digit and then represent them separately in binary 0111 1000.**

---

7. What are all the BCD numbers that can be uniquely represented in 2-4-2-1 weighted code?

**Answer:**

**0, 1, 8 and 9 (Only these 4 numbers will have unique representations).**

---

8. What is the condition for a weighted code to be self-complementary?

**Answer:**

**A weighted code is self-complimentary if the sum of the weights equals to 9. E.g.: 2-4-2-1 code. Sum of the weights =  $2+4+2+1 = 9$**

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